FlowState Ops

AI Event Congestion Decision Support System

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AI Event Congestion Decision Support System

- From Reactive Traffic Management to Predictive Traffic Operations
- Problem Theme: Event-Driven Congestion (Planned & Unplanned)
- Bengaluru Demo Scenario:
- 6 simultaneous city events overlapping:
- 🏏 IPL Match – Chinnaswamy Stadium
- 📢 Political Rally – Freedom Park
- 💻 Tech Summit – Palace Grounds
- 🎸 Music Festival – Whitefield
- 🏗 Trade Expo – BIEC
- 🛕 Temple Festival – Basavanagudi

One-Line Value Proposition:

- Predict congestion before it happens, recommend interventions, optimize resource deployment, and continuously learn from every event.

Why Current Traffic Management Fails

The Operational Challenge

- Every major city faces recurring congestion during political rallies, sports events, festivals, and sudden incidents.

The Current Process:

- Event Occurs ➔ Traffic Builds Up ➔ Authorities React ➔ Gridlock Already Happened

The 5 Key Failures of Today:

- Event impact is not quantified beforehand.
- Resource deployment is experience-driven (gut feeling).
- Diversion planning is largely manual.
- No post-event learning system exists.
- The same mistakes repeat every year.

The Status Quo Example:

- Event: IPL Match (40,000 attendees).
- Current System: Deploy officers based on what a commander remembers from last year.
- Result: Unpredictable congestion, over-exhausted police forces, and paralyzed CBD routes

FlowState Ops Architecture

From Data to Decisions

1. Data Inputs

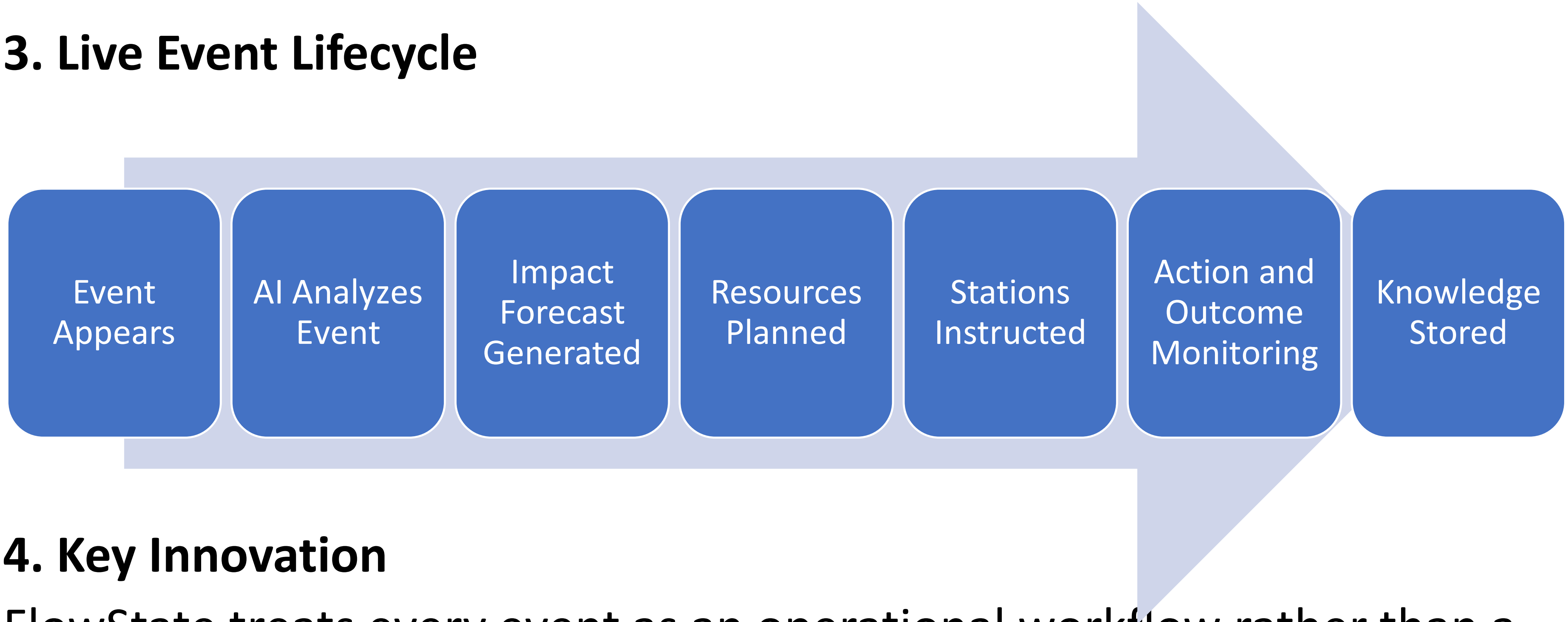
- Planned Events: Type, Location, Time, Expected Attendance
- Unplanned Events: Breakdowns, Roadblocks, Accidents
- Live Traffic Data: Speeds, Congestion levels, Corridor utilization
- Historical Archive: Previous actions, Outcomes, Lessons learned

2. Actionable Output

- Congestion Forecast & Delay Prediction
- Resource Plan (Officers, Barricades, Tows)
- Diversion Plan & Risk Assessment

FlowState Ops Architecture

3. Live Event Lifecycle



4. Key Innovation

FlowState treats every event as an operational workflow rather than a dashboard entry.

AI + ML + RAG Engine

- Under the Hood: Intelligence tied to Operations
- Here is how we translate raw machine learning into operational directives:

Component	Purpose (The Business Value)
Duration Predictor (LightGBM)	Predicts exactly how long the congestion will last.
Impact Model (LightGBM)	Predicts severity to establish priority level.
Geospatial Clustering (DBSCAN)	Detects physical hotspot corridors
Similar Event Retrieval (FAISS)	Gives the city a "memory" by finding historical matches.
Rule-Based Engine	Generates exact manpower and barricading requirements.
GenAI Explanation Layer	Translates complex model data into plain English for operators.

- GenAI Explanation: "91% similarity to IPL Final 2023. Diversion Route B reduced congestion by 41% in similar historical events."

Live Operations Dashboard

The Command Center

- Our UI is divided into four operational quadrants:

1. Event Intelligence (Left):

- Calculates the Event Impact Score, Forecasts Crowd sizes, and provides Historical Context.

2. Live Operations (Center):

- Dynamic Bengaluru Map, Congestion Heatmaps, and Active Incidents overlay.

3. Decision Support (Right):

- Precise Resource Planning, Diversion Plan execution, and a "What-If" Action Simulator.

4. Learning & KPIs (Bottom):

- Tracks Prediction Accuracy, Identifies Similar Events, and grades Playbook Improvements.

Demo Scenario

Watch the System in Action

The Situation:

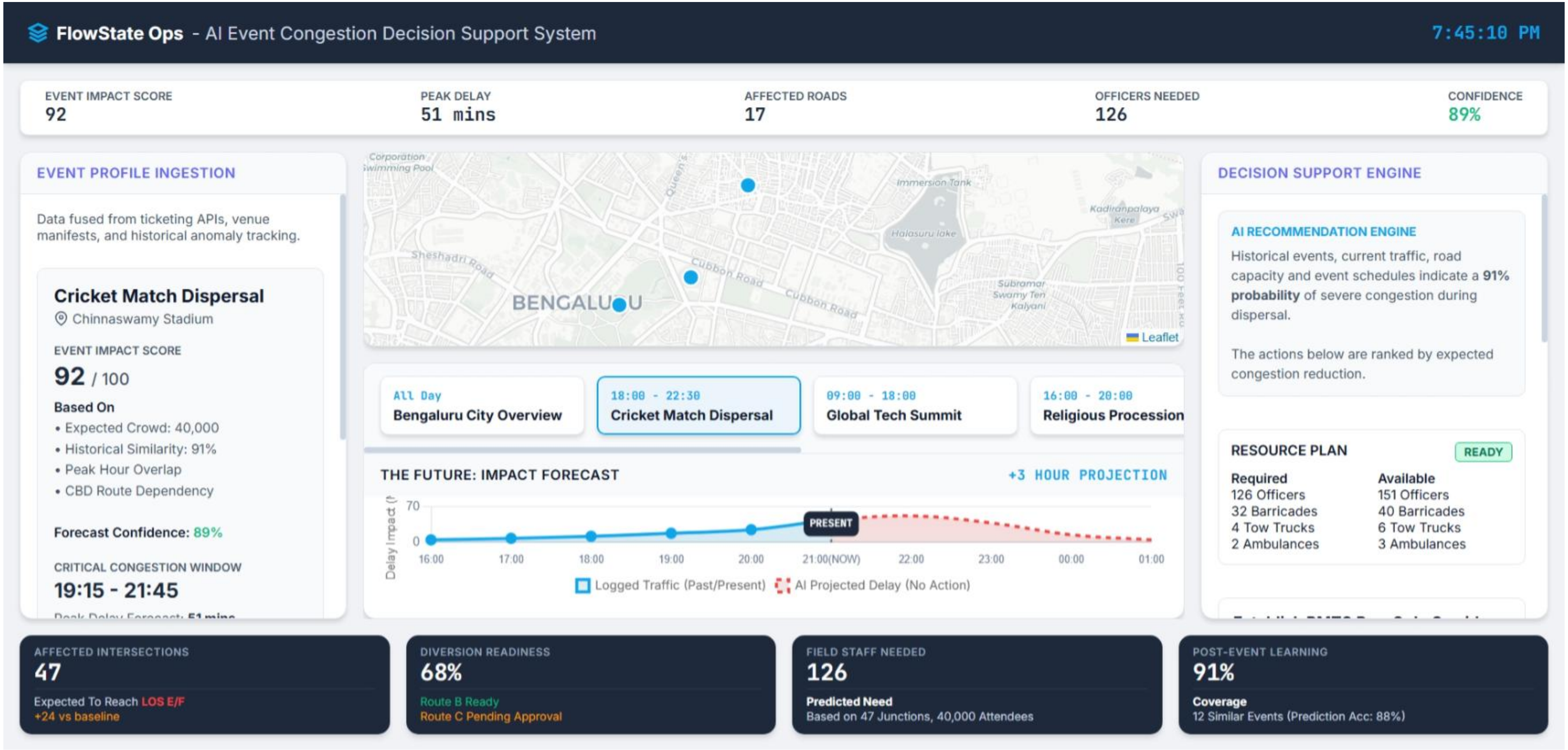
- An IPL Match and a Religious Procession overlap during evening peak hours.
- System Prediction (No Action):
- Congestion Score: 87 / 100
- Delay Expected: 51 minutes
- Collateral: 47 junctions affected

FlowState Recommended Actions:

- Deploy 24 Officers and 8 Barricades.
- Establish BMTC Bus-Only Corridor.
- Execute API Navigation Reroute away from MG Road.

The Simulation (The "What-If" Click):

- Without Action: 51 Min Delay
- With Action: 18 Min Delay
- Result: 64% Reduction in Gridlock.


FlowState Ops - AI Event Congestion Decision Support System
7:45:10 PM
EVENT IMPACT SCORE
92
PEAK DELAY
51 mins
AFFECTED ROADS
17
OFFICERS NEEDED
126
CONFIDENCE
89%
EVENT PROFILE INGESTION
Data fused from ticketing APIs, venue manifests, and historical anomaly tracking.
Cricket Match Dispersal
Chinnaswamy Stadium
EVENT IMPACT SCORE
92 / 100
Based On
• Expected Crowd: 40,000
• Historical Similarity: 91%
• Peak Hour Overlap
• CBD Route Dependency
Forecast Confidence: 89%
CRITICAL CONGESTION WINDOW
19:15 - 21:45
Corporation Swimming Pool
Immersion Tank
Halasuru lake
Kadiranpalaya Kere
Sheshadri Road
Cubbon Road
Cubbon Road
Subramar Swamy Ten Kalyani
BENGALURU
Leaflet
All Day
Bengaluru City Overview
18:00 - 22:30
Cricket Match Dispersal
09:00 - 18:00
Global Tech Summit
16:00 - 20:00
Religious Procession
THE FUTURE: IMPACT FORECAST
+3 HOUR PROJECTION
Delay Impact
70
0
PRESENT
16:00
17:00
18:00
19:00
20:00
21:00(NOW)
22:00
23:00
00:00
01:00
Logged Traffic (Past/Present)
AI Projected Delay (No Action)
DECISION SUPPORT ENGINE
AI RECOMMENDATION ENGINE
Historical events, current traffic, road capacity and event schedules indicate a 91% probability of severe congestion during dispersal.
The actions below are ranked by expected congestion reduction.
RESOURCE PLAN
READY
Required
126 Officers
32 Barricades
4 Tow Trucks
2 Ambulances
Available
151 Officers
40 Barricades
6 Tow Trucks
3 Ambulances
AFFECTED INTERSECTIONS
47
Expected To Reach LOS E/F
+24 vs baseline
DIVERSION READINESS
68%
Route B Ready
Route C Pending Approval
FIELD STAFF NEEDED
126
Predicted Need
Based on 47 Junctions, 40,000 Attendees
POST-EVENT LEARNING
91%
Coverage
12 Similar Events (Prediction Acc: 88%)

Impact & Future Vision

- **Predict ➔ Recommend ➔ Simulate ➔ Deploy ➔ Learn**
- Immediate Operational Benefits:
- Quantified event impact replacing guesswork.
- Optimized resource deployment (saving city budgets).
- Faster response times and significantly reduced congestion.
- Long-Term Civic Benefits:
- Creates a city-wide, continuously learning event intelligence platform.
- Reduces overall operational costs for traffic authorities.
- Massively improves the daily commuter experience.

The Roadmap:

- Phase 1 (MVP): Event forecasting & Resource recommendations (What we built today).
- Phase 2: Live integration with Google Maps APIs and local transit feeds.
- Phase 3: Autonomous API execution (e.g., adaptive signal control and automated VMS sign updates).

THANK YOU

-FlowState Ops transforms traffic management from a reactive scramble to a predictive science.